

An explicit obstruction to nuclear norm-ultrapower embeddings

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Abstract

We give a negative answer to Kirchberg’s norm-ultrapower embedding problem. We construct an explicit separable unital full group C^* -algebra that admits no unital embedding into B^ω , for any nonzero unital nuclear C^* -algebra B and any free ultrafilter ω on $\mathbb{N}$. In particular, it does not embed unitaly into $\mathcal{O}_2^\omega$.

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1 The embedding problem

Throughout, C^* -algebras are complex, and an *embedding* is an injective $*$ -homomorphism. We state unit preservation explicitly. For a nonzero unital C^* -algebra B and a free ultrafilter ω on $\mathbb{N}$, its *norm ultrapower* is

$$B^\omega = \ell^\infty(\mathbb{N}, B)/I_\omega(B), \quad I_\omega(B) = \{(b_n) : \lim_\omega \|b_n\| = 0\}.$$

Here $\ell^\infty(\mathbb{N}, B)$ is the algebra of bounded sequences, and the quotient uses the operator norm. The Cuntz algebra $\mathcal{O}_2$ is the universal unital C^* -algebra generated by two isometries s_1, s_2 satisfying $s_1 s_1^* + s_2 s_2^* = 1$; it was introduced by Cuntz [2].

The embedding theorem of Kirchberg–Phillips [8] states that every separable unital exact C^* -algebra embeds unitaly into $\mathcal{O}_2$. Here exactness means that minimal tensoring preserves short exact sequences. Passing from $\mathcal{O}_2$ to its norm ultrapower asks how far this universality extends beyond exact algebras. The techniques behind the embedding theorem led Kirchberg to consider this possibility, as recounted in [6, Section 1].

Kirchberg’s embedding problem asks whether every separable unital C^* -algebra admits a unital embedding into $\mathcal{O}_2^\omega$; see Goldbring–Sinclair [6, Section 3.1]. We give an explicit counterexample. The unital formulation is equivalent to the formulation without unit preservation [6, Section 3.1]. This question concerns norm ultrapowers of a nuclear C^* -algebra. It is distinct from the tensor-norm formulation of Kirchberg’s conjecture and from Connes’ embedding problem, as emphasized in [6, Section 1].

Theorem 1.1. *Let*

$$G = \mathbb{Z}[1/2]^3 \rtimes (\mathrm{SL}_3(\mathbb{Z}) \times \mathbb{Z}), \quad (M, k) \cdot v = 2^k Mv,$$

and let $A = C^(G)$ be its full group C^* -algebra. Then A is separable and unital. For every nonzero unital nuclear C^* -algebra B and every free ultrafilter ω on $\mathbb{N}$, there is no unital embedding*

$$A \longrightarrow B^\omega.$$

In particular, A does not embed unitaly into $\mathcal{O}_2^\omega$.

The group is countable and independent of both B and ω ; no separability assumption on B is needed. The proof treats an arbitrary fixed nuclear B throughout. Specializing to the nuclear algebra $\mathcal{O}_2$ gives the negative answer to the embedding problem; for its nuclearity, see [11, Section 4, pp. 80–82]. Section 6 gives the consequences for existential closure, permanence of $\mathcal{O}_2^\omega$ -embeddability under crossed products, and good nuclear witnesses, with the relevant notions defined there.

1.1 Previous work and the new obstruction

Recent approaches study computability and tensor norms. Fox–Goldbring–Hart [5] derive computability-theoretic consequences of a positive answer, using the computable presentation of $\mathcal{O}_2$. Goldbring–Sinclair give a conditional route to a negative answer through a *weakly explicit minimal tensor norm*: an effective choice of universal formulas approximating the minimal tensor-norm predicates to a fixed accuracy. They leave this property of $\mathcal{O}_2$ open [7, Definition 4.16, Theorem 4.18 and Question 4.20]. Our obstruction instead comes from the representation theory of an explicit discrete group.

The group configuration already appears in the nonfiniteness arguments of Sauers [13, Section 2.4] and Eckhardt [4, Theorem 3.1 and Example 3.2]. They show that the full algebra of the ascending HNN extension G is nonfinite: it contains a proper isometry v , so $v^*v = 1$ but $vv^* \neq 1$. Its property (T) base group $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and its finite-index scalar self-embeddings are recorded in de Cornulier [3, Proposition 1.1 and its proof]. Sauers and Eckhardt also construct non-MF groups: groups admitting no injective homomorphism into the unitary group of any norm matrix corona $\prod_j M_{d_j} / \bigoplus_j M_{d_j}$, with positive integer dimensions d_j . Here the product consists of bounded sequences and the direct sum is the ideal of sequences tending to zero in norm [4, Theorem 4.3][13, Sections 3–5]. Eckhardt’s example is an associated wreath product. The target class here is different: nonfiniteness of $C^*(G)$ alone cannot exclude a unital embedding into $\mathcal{O}_2^\omega$, which itself contains proper isometries.

The additional ingredient is a finiteness statement for approximate finite-dimensional representation types inside a nuclear algebra (Lemma 4.3). Its tensor-product argument follows Wassermann's method [16], as explained by Manuilov–Thomsen [10, Section 2.1]. The statement applies to an arbitrary tuple of unitaries, even when the tuple satisfies no group relations. This flexibility is essential because coordinate lifts of an ultrapower representation are only approximately multiplicative. A dimension-independent estimate allows the representation type to vary with the coordinate. Finite-index induction then relates members of the resulting finite sets by composition with the self-embedding. These two passages turn the group configuration into the norm-ultrapower obstruction.

1.2 Proof outline

The subgroup $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ of G has property (T). Conjugation by the generator t of the last $\mathbb{Z}$ -factor sends Γ onto the index-eight subgroup $\theta(\Gamma)$, where $\theta(v, M) = (2v, M)$. There is a finite-dimensional irreducible representation π_* of Γ such that $\pi_* \circ \theta^j$ is inequivalent to π_* for every $j \geq 1$. Section 2 verifies these facts.

Fix a finite generating set S of Γ for which property (T) gives a spectral gap. Suppose a unital embedding $A \rightarrow B^\omega$ exists, and choose unitary lifts $x_n(g) \in B$ of its group unitaries. For each n , consider the nonzero finite-dimensional irreducible unitary representations π for which the tuple $(x_n(s) \otimes \bar{\pi}(s))_{s \in S}$ has spectral minimum below a fixed small threshold for the sum of squared displacements from 1. Here $\bar{\pi}$ is the conjugate representation. Sections 3 and 4 specify the threshold and show that the resulting set D_n of representation classes is finite.

Write $\pi \prec \rho$ when π occurs as a unitary subrepresentation of ρ , and call π' a predecessor of π if $\pi \prec \pi' \circ \theta$. Faithfulness of the embedding places $[\pi_*]$ in D_n on a set belonging to ω . In Section 5, the spectral gap for *every sequence* of finite-dimensional representations supplies, on one set belonging to ω , the simultaneous implication

$$[\pi] \in D_n \implies \text{some } [\pi'] \in D_n \text{ satisfies } \pi \prec \pi' \circ \theta.$$

Starting at π_* and repeatedly choosing a predecessor must give a repetition in the finite set D_n . Irreducibility then makes π_* periodic under composition with θ , a contradiction.

2 The group and a finite-dimensional representation

We construct a finite-dimensional representation inequivalent to each of its positive pullbacks along a fixed group endomorphism. The same group configuration appears in the nonfiniteness arguments of Sauers [13, Section 2.4] and Eckhardt [4, Theorem 3.1 and Example 3.2]. De Cornulier [3, Proposition 1.1 and its proof] records the property (T) affine group and its finite-index scalar self-embeddings. We use the dyadic case.

We use column vectors. In the group G of Theorem 1.1, write elements as (v, M, k) , so that

$$(v, M, k)(w, N, \ell) = (v + 2^k Mw, MN, k + \ell).$$

Set

$$\Gamma = \{(v, M, 0) : v \in \mathbb{Z}^3, M \in \mathrm{SL}_3(\mathbb{Z})\}, \quad t = (0, I, 1).$$

We suppress the final coordinate when writing elements of Γ , and define

$$\theta : \Gamma \longrightarrow \Gamma, \quad \theta(v, M) = (2v, M), \quad \Lambda = \theta(\Gamma).$$

Lemma 2.1. *The group Γ is finitely generated and has property (T). The map θ is injective,*

$$[\Gamma : \Lambda] = 8, \quad t\gamma t^{-1} = \theta(\gamma) \quad (\gamma \in \Gamma).$$

There is a nonzero finite-dimensional irreducible unitary representation π_ of Γ that factors through reduction modulo 2 and is nontrivial on the translation subgroup $\mathbb{Z}^3$. In particular,*

$$\pi_* \not\cong \pi_* \circ \theta^j \quad (j \geq 1). \quad (1)$$

Proof. Property (T) is the case $n = 3$, $m = 1$ of [3, Proposition 1.1], with $\mathrm{SL}_3(\mathbb{Z})$ acting on one copy of its standard rank-three integral module. For finite generation, take the three standard translations and the elementary matrices $I + E_{ij}$, $i \neq j$, where E_{ij} is the matrix unit. These matrices generate $\mathrm{SL}_3(\mathbb{Z})$ by integer row reduction.

The group law shows that θ is an injective homomorphism and that conjugation by t implements it. Its image is $\Lambda = 2\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$. The elements

$$r_a = (a, I), \quad a \in \{0, 1\}^3,$$

are representatives of its eight left cosets: writing $v = a + 2w$ gives $(v, M) = r_a(2w, M)$, and two such representatives give the same coset only when they have the same residue modulo 2.

Let F be the finite image of the reduction homomorphism

$$\Gamma \longrightarrow (\mathbb{Z}/2\mathbb{Z})^3 \rtimes \mathrm{SL}_3(\mathbb{Z}/2\mathbb{Z}), \quad (v, M) \longmapsto (v \bmod 2, M \bmod 2).$$

The translation subgroup $(\mathbb{Z}/2\mathbb{Z})^3$ acts nontrivially in the left regular representation of F . Decomposing this representation into irreducibles therefore gives an irreducible constituent that is nontrivial on translations. Its pullback to Γ is the required π_* . Indeed, π_* factors through reduction modulo 2, whereas $\theta^j(v, I) = (2^j v, I)$. Thus $\pi_* \circ \theta^j$ is trivial on all translations for every $j \geq 1$, and π_* is not. This proves (1). $\square$

The group G is countable, so its full group C^* -algebra $C^*(G)$ is separable and unital. Every unitary representation of G extends to $C^*(G)$. We will use this universal property after inducing π_* from Γ to G .

3 A spectral test for representation matching

We now introduce the spectral quantity used to form the finite sets in the proof outline. The discussion applies to any finitely generated discrete group Γ with property (T). Choose a nonempty finite generating set S containing a Kazhdan set. By property (T), after decreasing the constant if necessary, there is $0 < \kappa \leq 4|S|$ such that every unitary representation σ with no nonzero invariant vector satisfies

$$\sum_{s \in S} \|\sigma(s)\xi - \xi\|^2 \geq \kappa \|\xi\|^2. \quad (2)$$

For a tuple $\mathbf{z} = (z_s)_{s \in S}$ of unitaries in a nonzero unital C^* -algebra C , set

$$L(\mathbf{z}) = \sum_{s \in S} (z_s - 1)^*(z_s - 1), \quad R = 4|S|. \quad (3)$$

Thus $0 \leq L(\mathbf{z}) \leq R1$, without any group relations on $\mathbf{z}$.

Lemma 3.1. *If $z_s = \sigma(s)$ for a unitary representation $\sigma : \Gamma \rightarrow \mathcal{U}(C)$ in a nonzero unital C^* -algebra, then*

$$\text{spec } L(\mathbf{z}) \subseteq \{0\} \cup [\kappa, R].$$

Proof. Represent C faithfully and unitaly on a Hilbert space. The invariant vectors for σ and their orthogonal complement are reducing subspaces. On the former L vanishes; on the latter (2) gives $L \geq \kappa 1$. A faithful unital representation preserves spectra, proving the claim. $\square$

Every finite-dimensional representation below acts on a nonzero Hilbert space. If $\pi : \Gamma \rightarrow \mathcal{U}(V_\pi)$ is such a representation, its conjugate $\bar{\pi}$ acts on the conjugate Hilbert space $\bar{V}_\pi$ by $\bar{\pi}(g)\bar{v} = \overline{\pi(g)v}$. Define

$$d(\mathbf{z}, \pi) = \min \text{spec } L((z_s \otimes \bar{\pi}(s))_{s \in S}), \quad (4)$$

where the spectrum is taken in $C \otimes \mathcal{B}(\bar{V}_\pi)$. Matrix algebras have a unique C^* -tensor norm. On any faithful unital Hilbert-space realization $C \subseteq \mathcal{B}(H)$, the number in (4) equals

$$\inf_{\substack{\xi \in H \otimes \bar{V}_\pi \\ \|\xi\|=1}} \sum_{s \in S} \|(z_s \otimes \bar{\pi}(s))\xi - \xi\|^2.$$

In such a faithful realization, if $z_s = \rho(s)$ comes from a unitary representation and π is irreducible, then $d(\mathbf{z}, \pi) = 0$ exactly when π occurs in ρ , and otherwise $d(\mathbf{z}, \pi) \geq \kappa$. Indeed, invariant vectors of $\rho \otimes \bar{\pi}$ correspond to intertwiners from V_π to H , and the spectral gap separates their orthogonal complement from zero.

The number $d(\mathbf{z}, \pi)$ is unchanged by unitary equivalence of π . For a finite direct sum, the same formula gives

$$d(\mathbf{z}, \pi_1 \oplus \cdots \oplus \pi_r) = \min_{1 \leq i \leq r} d(\mathbf{z}, \pi_i). \quad (5)$$

Fix once and for all

$$b = \frac{\kappa^2}{4|S|}, \quad 0 < a < \min\{\kappa, b\}. \quad (6)$$

The next section proves that, for a tuple in a nuclear algebra, only finitely many irreducible π can satisfy $d(\mathbf{z}, \pi) < a$. No group relations on that tuple will be needed.

4 Finiteness of approximate representation matches

We prove that, for a fixed unitary tuple $\mathbf{u}$ in a nuclear algebra, only finitely many finite-dimensional irreducible representation classes satisfy $d(\mathbf{u}, \pi) < a$. The tuple need not satisfy any group relations. This distinction will allow us to apply the result to coordinate lifts from a norm ultrapower.

We retain the notation and constants of Section 3. For C^* -algebras B and C , the minimal tensor product $B \otimes_{\min} C$ is realized by faithful representations on separate Hilbert spaces; the maximal tensor product $B \otimes_{\max} C$ is characterized by pairs of commuting representations on one Hilbert space. We use the tensor-norm characterization of nuclearity developed by Takesaki and Lance [14, 9]: B is nuclear if these two tensor norms agree for every C . We first obtain a lower bound when the second tensor factor has no finite-dimensional subrepresentations. The passage from a vector to its reduced density operator removes the arbitrary unitaries in the first factor.

Lemma 4.1 (Square-root trace inequality). *For positive Hilbert–Schmidt operators X, Y on any Hilbert space,*

$$\|X - Y\|_2^2 \leq \|X^2 - Y^2\|_1.$$

Here $\|\cdot\|_1$ and $\|\cdot\|_2$ denote the trace and Hilbert–Schmidt norms, respectively.

Proof. This is the square-root trace inequality of Powers–Størmer [12, Lemma 4.1]; we include the short argument needed here. Put $F = X - Y$, and let $J = \text{sgn}(F)$, with value zero on $\ker F$. Since $X^2 - Y^2 = FX + YF$ and $JF = FJ = |F|$,

$$\begin{aligned}\|X^2 - Y^2\|_1 &\geq \text{Tr}(J(X^2 - Y^2)) \\ &= \text{Tr}(|F|(X + Y)) \\ &\geq \text{Tr}(F_+F) + \text{Tr}(F_-(-F)) = \text{Tr}(F^2).\end{aligned}$$

For the last inequality, use $X + Y - F = 2Y \geq 0$ and $X + Y + F = 2X \geq 0$, and pair these inequalities with F_+ and F_- , respectively.

All these traces are well defined in infinite dimension. Products of two Hilbert–Schmidt operators are trace class; their traces are cyclic, by finite-rank approximation and $\|AB\|_1 \leq \|A\|_2\|B\|_2$. The trace of a product of two positive Hilbert–Schmidt operators is nonnegative, as is seen by diagonalizing one factor. These observations also justify the cyclic rearrangement with the bounded operator J . The operators involved vanish on the orthogonal complement of a separable subspace, so no separability hypothesis on the ambient Hilbert space is needed. $\square$

Bekka [1] proved that if a unitary representation of a property (T) group on a Hilbert space K admits a state on $\mathcal{B}(K)$ invariant under conjugation by the representation, then it has a nonzero finite-dimensional subrepresentation; see also [15, proof of Proposition 2]. The next lemma gives the explicit spectral bound needed here for an arbitrary unitary tuple in the first tensor factor.

Lemma 4.2 (Uniform spectral bound). *Let Γ, S, κ satisfy (2). Let $\sigma : \Gamma \rightarrow \mathcal{U}(K)$ be a unitary representation with no nonzero finite-dimensional subrepresentation. For any Hilbert space E and any unitaries $(U_s)_{s \in S}$ on E ,*

$$L((U_s \otimes \sigma(s))_{s \in S}) \geq b1, \quad b = \frac{\kappa^2}{4|S|}.$$

No relations among the U_s are required.

Proof. Conjugation by σ is a unitary representation on the Hilbert space $\mathcal{S}_2(K)$ of Hilbert–Schmidt operators. It has no nonzero invariant vector: if $A \neq 0$ commutes with every $\sigma(g)$, then so does the nonzero positive compact operator $|A|$. The eigenspace of any nonzero eigenvalue of $|A|$ is a nonzero finite-dimensional σ -invariant subspace.

Use inner products linear in the first variable. Fix a unit vector $\xi \in E \otimes K$. Its reduced density operator T on K is characterized by

$$\text{Tr}(TC) = \langle (1 \otimes C)\xi, \xi \rangle \quad (C \in \mathcal{B}(K)).$$

Concretely, write $\xi = \sum_i e_i \otimes v_i$ with (e_i) orthonormal and $\sum_i \|v_i\|^2 = 1$. Then $T = \sum_i |v_i\rangle\langle v_i|$, with convergence in trace norm; in particular, $T \geq 0$ and $\text{Tr}(T) = 1$. Such an expansion uses only countably many vectors, even if the Hilbert spaces are nonseparable.

For $s \in S$, put $\xi_s = (U_s \otimes \sigma(s))\xi$ and $\delta_s = \|\xi_s - \xi\|$. The reduced density operator of ξ_s is $T_s = \sigma(s)T\sigma(s)^*$. Testing the two vector states against $1 \otimes C$, for $\|C\| \leq 1$, gives

$$\|T_s - T\|_1 \leq 2\delta_s.$$

Lemma 4.1 therefore gives

$$\|\sigma(s)T^{1/2}\sigma(s)^* - T^{1/2}\|_2^2 \leq \|T_s - T\|_1 \leq 2\delta_s.$$

Since $\|T^{1/2}\|_2 = 1$, applying (2) to the conjugation representation and then Cauchy–Schwarz yields

$$\kappa \leq 2 \sum_{s \in S} \delta_s \leq 2\sqrt{|S|} \left(\sum_{s \in S} \delta_s^2 \right)^{1/2}.$$

Thus $\sum_s \delta_s^2 \geq b$. Taking all unit vectors ξ proves the operator inequality; when $E \otimes K = \{0\}$ it is vacuous. $\square$

The following quotient argument uses Wassermann’s property (T) method [16]; see the exposition in [10, Section 2.1]. The preceding uniform bound allows us to use an arbitrary unitary tuple in the first factor. Nuclearity will transfer the bound to the coordinate quotient and contradict the assumed small spectral minima.

Lemma 4.3 (Nuclear finiteness). *Let $\Gamma, S, \kappa, R, b, a$ be as above. For every nonzero unital nuclear C^* -algebra B and every unitary S -tuple $\mathbf{u}$ in B , the set*

$$D(\mathbf{u}) = \{[\pi] : d(\mathbf{u}, \pi) < a\}$$

is finite, where π ranges over the nonzero finite-dimensional irreducible unitary representations of Γ and $[\pi]$ denotes its unitary equivalence class. No group relations on $\mathbf{u}$ are assumed.

Proof. Suppose that $(\pi_j)_{j \geq 1}$ are pairwise inequivalent representations with $[\pi_j] \in D(\mathbf{u})$. Write $M_j = \mathcal{B}(\bar{V}_{\pi_j})$, and form

$$Q = \frac{\prod_{j \geq 1} M_j}{\bigoplus_{j \geq 1} M_j}, \quad q(g) = [(\bar{\pi}_j(g))_j] \quad (g \in \Gamma).$$

The product consists of bounded sequences, and the direct sum is the ideal of sequences whose norms tend to zero in the ordinary sense. In particular, this auxiliary quotient does not involve an ultrafilter.

Choose a faithful unital representation $\eta : Q \rightarrow \mathcal{B}(K)$, and put $\sigma = \eta \circ q$. We first show that σ has no nonzero finite-dimensional subrepresentations. Fix a finite-dimensional irreducible representation τ of Γ . Under the identification

$$\bar{V}_{\pi_j} \otimes \bar{V}_\tau \simeq \mathcal{S}_2(V_\tau, \bar{V}_{\pi_j}),$$

invariant vectors for $\bar{\pi}_j \otimes \bar{\tau}$ correspond to intertwiners from τ to $\bar{\pi}_j$. By irreducibility and inequivalence, such an intertwiner can exist for at most one j . Consequently,

$$L((\bar{\pi}_j(s) \otimes \bar{\tau}(s))_{s \in S}) \geq \kappa 1$$

outside at most one coordinate. Fixed-size matrix amplification commutes with this quotient, entry by entry:

$$Q \otimes \mathcal{B}(\bar{V}_\tau) \simeq \frac{\prod_j (M_j \otimes \mathcal{B}(\bar{V}_\tau))}{\bigoplus_j (M_j \otimes \mathcal{B}(\bar{V}_\tau))}.$$

The exceptional coordinate disappears, and hence

$$L((q(s) \otimes \bar{\tau}(s))_{s \in S}) \geq \kappa 1. \tag{7}$$

If σ contained τ , then $\sigma \otimes \bar{\tau}$ would have an invariant vector, contradicting (7). Every nonzero finite-dimensional unitary representation contains an irreducible subrepresentation, so σ has none.

Represent B faithfully and unitaly on a Hilbert space E . By Lemma 4.2,

$$L((u_s \otimes q(s))_{s \in S}) \geq b1 \quad \text{in } B \otimes_{\min} Q. \quad (8)$$

We now use nuclearity to pass this inequality to the coordinate quotient. Let

$$C = \frac{\prod_j (B \otimes M_j)}{\bigoplus_j (B \otimes M_j)}.$$

There are commuting unital $*$ -homomorphisms

$$\alpha : B \longrightarrow C, \quad \alpha(x) = [(x \otimes 1_{M_j})_j], \quad \beta : Q \longrightarrow C, \quad \beta([(y_j)_j]) = [(1_B \otimes y_j)_j].$$

The map β is well defined because $\|1_B \otimes y_j\| = \|y_j\|$. The universal property of the maximal tensor product gives a unital $*$ -homomorphism

$$\Psi : B \otimes_{\max} Q \longrightarrow C, \quad \Psi(x \otimes q) = \alpha(x)\beta(q).$$

Since B is nuclear, $B \otimes_{\min} Q = B \otimes_{\max} Q$, so (8) passes through Ψ .

Put $L_j = L((u_s \otimes \pi_j(s))_{s \in S})$. We have proved $[(L_j)_j] \geq b1$ in C . Continuous functional calculus for $f(t) = (b - t)_+$ now gives

$$[(f(L_j))_j] = f([(L_j)_j]) = 0, \quad \|f(L_j)\| = (b - d(\mathbf{u}, \pi_j))_+ \longrightarrow 0.$$

Thus $\liminf_j d(\mathbf{u}, \pi_j) \geq b$, whereas every term is less than $a < b$. This contradiction proves finiteness. $\square$

The only use of nuclearity was to identify the minimal and maximal tensor products before applying Ψ . In particular, the proof does not assume an identification of the minimal tensor product with the coordinate quotient, and it imposes no bound on the dimensions of the π_j .

5 From an ultrapower embedding to a finite contradiction

To prove Theorem 1.1, fix a nonzero unital nuclear C^* -algebra B and a free ultrafilter ω on $\mathbb{N}$. A set of indices is called ω -large if it belongs to ω . Suppose, towards a contradiction, that there is a unital injective $*$ -homomorphism

$$\Phi : C^*(G) \longrightarrow B^\omega.$$

We will associate a finite set of irreducible representations of Γ to each coordinate of Φ . Conjugation by t will force these sets to contain predecessors under restriction by θ . The resulting finite chains contradict the choice of π_* .

5.1 Unitary lifts and a gap for varying matrix dimensions

Write u_g for the canonical unitary of $g \in G$ in $C^*(G)$. Each unitary in B^ω has a representative consisting of unitaries in B . Indeed, if (y_n) represents a unitary, then

$$\lim_{\omega} \|y_n^* y_n - 1\| = \lim_{\omega} \|y_n y_n^* - 1\| = 0.$$

On an ω -large set both norms are less than $1/2$, so y_n is invertible and $y_n(y_n^* y_n)^{-1/2}$ is unitary. The difference between this unitary and y_n tends to zero along ω , by continuous functional calculus. Set the representative equal to 1 outside this set.

Choose such representatives $x_n(g) \in \mathcal{U}(B)$ of $\Phi(u_g)$ for every $g \in G$, with $x_n(e) = 1$. For each fixed pair $g, h \in G$,

$$\lim_{\omega} \|x_n(g)x_n(h) - x_n(gh)\| = 0. \quad (9)$$

These choices impose no group relations at any individual coordinate. All uses of (9) below involve finitely many fixed group elements; no intersection over all pairs is needed. Put $\mathbf{x}_n = (x_n(s))_{s \in S}$. Among the unitary equivalence classes of nonzero finite-dimensional irreducible representations of Γ , define

$$D_n = \{[\pi] : d(\mathbf{x}_n, \pi) < a\}. \quad (10)$$

Here brackets denote unitary equivalence classes. Each D_n is finite by Lemma 4.3, which applies to arbitrary unitary S -tuples.

The next observation provides uniform control even when the dimensions of the representations increase with n .

Lemma 5.1. *For every sequence (π_n) of nonzero finite-dimensional unitary representations of Γ , with no bound on their dimensions,*

$$\lim_{\omega} d(\mathbf{x}_n, \pi_n) \in \{0\} \cup [\kappa, R]. \quad (11)$$

In particular, if $\{n : d(\mathbf{x}_n, \pi_n) < a\} \in \omega$, then this limit is zero.

Proof. Let $C_n = B \otimes \mathcal{B}(\overline{V_{\pi_n}})$, and form the norm ultraproduct

$$C_{\omega} = \prod_n C_n / \{(c_n) : \lim_{\omega} \|c_n\| = 0\}.$$

The product consists of bounded sequences. For every such sequence,

$$\|[(c_n)]\| = \lim_{\omega} \|c_n\|. \quad (12)$$

For the lower bound, adding an ω -null sequence does not change the ultralimit of the norms. For the upper bound, if this ultralimit is c , replace c_n by 0 outside $\{n : \|c_n\| < c + \varepsilon\} \in \omega$ and then let $\varepsilon \downarrow 0$.

Equation (9) shows that

$$g \mapsto [(x_n(g) \otimes \overline{\pi_n}(g))_n], \quad g \in \Gamma,$$

is a genuine unitary representation in the nonzero unital algebra C_{ω} . By Lemma 3.1, its Laplacian has spectrum in $\{0\} \cup [\kappa, R]$. Write

$$P_n = L((x_n(s) \otimes \overline{\pi_n}(s))_{s \in S}).$$

Since $0 \leq P_n \leq R1$, equation (12) gives

$$\begin{aligned} \min \text{spec}[(P_n)] &= R - \|R1 - [(P_n)]\| \\ &= \lim_{\omega} (R - \|R1 - P_n\|) = \lim_{\omega} d(\mathbf{x}_n, \pi_n). \end{aligned}$$

This proves (11). In the final assertion the limit is at most $a < \kappa$, and hence must be zero. $\square$

5.2 The initial representation occurs

We next use injectivity of Φ to ensure that $[\pi_*] \in D_n$ for an ω -large set of indices.

Lemma 5.2. *We have $\lim_{\omega} d(\mathbf{x}_n, \pi_*) = 0$, and consequently $\{n : [\pi_*] \in D_n\} \in \omega$.*

Proof. Induce π_* from Γ to G on the Hilbert direct sum of copies of V_{π_*} indexed by the left cosets G/Γ . Explicitly, choose representatives q_i , including the identity: when $gq_i = q_j\gamma$ with $\gamma \in \Gamma$, the action sends coordinate i to coordinate j by $\pi_*(\gamma)$. This defines a unitary representation. The identity coordinate is invariant under Γ and carries π_* . Thus the restriction of this representation of G contains π_* . Tensoring with $\overline{\pi_*}$ gives a nonzero Γ -invariant vector: on this coordinate it is $\sum_j e_j \otimes \overline{e_j}$ for an orthonormal basis (e_j) of V_{π_*} .

Every unitary representation of G gives a representation of the full group algebra $C^*(G)$. The preceding invariant vector shows that

$$\min \text{spec } L((u_s \otimes \overline{\pi_*}(s))_{s \in S}) = 0.$$

Indeed, an invertible positive element could not acquire a kernel under a unital representation. The matrix amplification of Φ is injective and preserves spectra, so the corresponding spectral minimum in $B^{\omega} \otimes \mathcal{B}(\overline{V_{\pi_*}})$ is also zero.

For each fixed positive integer d there is the canonical identification

$$M_d(B^{\omega}) \cong (M_d(B))^{\omega}.$$

To check its kernel, a sequence of d -by- d matrices tends to zero in norm along ω precisely when each of its finitely many entries does. Entrywise lifts give surjectivity. Applying this identification with $d = \dim \pi_*$ and the calculation of spectral minima in Lemma 5.1 proves the assertion. This fixed matrix amplification is distinct from the varying algebras used in that lemma. $\square$

5.3 A predecessor for every representation at one coordinate

For representations α and β of the same group, write $\alpha \prec \beta$ when α is unitarily equivalent to a subrepresentation of β . The required quantifier order is the following: on one ω -large set, every member of D_n has a predecessor in that same finite set.

Lemma 5.3. *The set*

$$\{n : (\forall [\pi] \in D_n)(\exists [\pi'] \in D_n) \quad \pi \prec \pi' \circ \theta\}$$

belongs to ω .

Proof. Suppose otherwise. On an ω -large set F , choose an irreducible π_n with $[\pi_n] \in D_n$ for which no such predecessor exists. Outside F , choose the trivial representation. By Lemma 5.1,

$$\lim_{\omega} d(\mathbf{x}_n, \pi_n) = 0. \tag{13}$$

Set $\alpha_n = \pi_n \circ \theta^{-1}$, a representation of Λ , and let $\rho_n = \text{Ind}_{\Lambda}^{\Gamma} \alpha_n$. We will prove that $\lim_{\omega} d(\mathbf{x}_n, \rho_n) = 0$ and that every irreducible constituent π' of ρ_n satisfies $\pi_n \prec \pi' \circ \theta$. The block-minimum identity (5) will then select a constituent belonging to D_n on an ω -large set, and the second assertion will make that constituent a predecessor of π_n .

Transport to Λ . Represent B faithfully and unitaly on a Hilbert space $\mathcal{H}$, and use the same notation for its unitaries on $\mathcal{H}$. Choose unit vectors $\xi_n \in \mathcal{H} \otimes \overline{V_{\pi_n}}$ such that

$$\sum_{s \in S} \|(x_n(s) \otimes \overline{\pi_n}(s))\xi_n - \xi_n\|^2 < d(\mathbf{x}_n, \pi_n) + 1/n.$$

Because ω is free, $1/n$ tends to zero along ω . Thus

$$\lim_{\omega} \|(x_n(g) \otimes \overline{\pi_n}(g))\xi_n - \xi_n\| = 0 \quad (g \in \Gamma). \quad (14)$$

Here is the fixed-word justification. Put $U_n(g) = x_n(g) \otimes \overline{\pi_n}(g)$. For each $s \in S$, $x_n(s^{-1}) - x_n(s)^*$ tends to zero by (9); a unitary and its inverse have the same displacement norm on a vector. For a fixed word $g = s_1^{\varepsilon_1} \cdots s_k^{\varepsilon_k}$, repeated use of (9) replaces $U_n(g)$ by the corresponding product of $U_n(s_j)^{\varepsilon_j}$ with an error tending to zero. The displacement of that product is at most the sum of the k displacements of its factors. Both the word and k are fixed while n varies.

Put $\eta_n = (x_n(t) \otimes 1)\xi_n$. These are unit vectors. For $h \in \Lambda$ put $g = \theta^{-1}(h)$. The identity $ht = tg$ and (9) give

$$\lim_{\omega} \|x_n(h)x_n(t) - x_n(t)x_n(g)\| = 0.$$

Together with (14), this proves

$$\lim_{\omega} \|(x_n(h) \otimes \overline{\alpha_n}(h))\eta_n - \eta_n\| = 0 \quad (h \in \Lambda). \quad (15)$$

Induction to Γ . We transfer these vectors from Λ to Γ by a fixed eight-coordinate induction. Use the left coset representatives $r_1 = e, r_2, \dots, r_8$ of Γ/Λ given by the translations in $\{0, 1\}^3$. For $g \in \Gamma$ and $1 \leq i \leq 8$, define $j(g, i)$ and $h(g, i) \in \Lambda$ uniquely by

$$gr_i = r_{j(g,i)}h(g, i).$$

The induced representation ρ_n acts on $\bigoplus_{i=1}^8 V_{\pi_n}$: $\rho_n(g)$ sends coordinate i to coordinate $j(g, i)$ by $\alpha_n(h(g, i))$. The relation

$$h(g_1 g_2, i) = h(g_1, j(g_2, i))h(g_2, i)$$

verifies the representation law. Permutation of the coordinates and unitarity of α_n verify unitarity. Its conjugate has the same coordinate rule with $\overline{\alpha_n}$ in place of α_n .

Define $\zeta_n \in \mathcal{H} \otimes \overline{V_{\rho_n}}$ by

$$(\zeta_n)_i = 8^{-1/2}(x_n(r_i) \otimes 1)\eta_n.$$

The eight summands are orthogonal and each has norm $8^{-1/2}$, so $\|\zeta_n\| = 1$. Fix $s \in S$ and i , and write $j = j(s, i)$ and $h = h(s, i)$. The component sent from i to j by $x_n(s) \otimes \overline{\rho_n}(s)$ differs from $(\zeta_n)_j$ by a vector whose norm is at most

$$8^{-1/2}(\|x_n(s)x_n(r_i) - x_n(r_j)x_n(h)\| + \|(x_n(h) \otimes \overline{\alpha_n}(h))\eta_n - \eta_n\|).$$

The first term tends to zero by (9) and $sr_i = r_j h$; the second does so by (15). There are only $8|S|$ fixed choices involved. Hence

$$\lim_{\omega} \sum_{s \in S} \|(x_n(s) \otimes \overline{\rho_n}(s))\zeta_n - \zeta_n\|^2 = 0.$$

Thus $\lim_{\omega} d(\mathbf{x}_n, \rho_n) = 0$, and in particular $d(\mathbf{x}_n, \rho_n) < a$ on an ω -large set. All error bounds are independent of $\dim \pi_n$.

Selecting a predecessor. Decompose the finite-dimensional unitary representation ρ_n into irreducible summands. The corresponding Laplacian is a direct sum, so its spectral minimum is the minimum of the spectral minima of these summands. Consequently, on an ω -large set, some irreducible constituent π'_n of ρ_n has $[\pi'_n] \in D_n$.

Every irreducible constituent π' of ρ_n , including the one just obtained, satisfies $\pi_n \prec \pi' \circ \theta$. This is the finite-index form of Frobenius reciprocity. To verify it for every constituent, let W be its invariant subspace and let $P_1 : W \rightarrow V_{\pi_n}$ be projection onto the identity coordinate. This map is Λ -equivariant for the target representation α_n . It is nonzero: if $w \in W$ has a nonzero i th coordinate, then

$$P_1 \rho_n(r_i^{-1})w = w_i \neq 0.$$

Since α_n is irreducible, $P_1|_W$ is surjective. Its adjoint is an injective Λ -intertwiner; after normalization by a positive scalar it is an isometric intertwiner, because $(P_1|_W)(P_1|_W)^*$ commutes with the irreducible α_n . It follows that $\alpha_n \prec \pi'|_\Lambda$; composing with $\theta : \Gamma \rightarrow \Lambda$ gives the asserted containment.

Intersect the ω -large set on which such a $[\pi'_n]$ belongs to D_n with F . This contradicts the choice of π_n and proves the lemma. $\square$

5.4 Finite chains force periodicity

The analytic argument is complete. We finish with the elementary consequence of finiteness that makes the predecessor property incompatible with π_* . Consider a chain of finite-dimensional irreducible unitary representations $\tau_0 = \pi, \tau_1, \dots$ of Γ whose classes lie in a finite set D and satisfy $\tau_i \prec \tau_{i+1} \circ \theta$. Finiteness gives indices $l \geq 0$ and $p \geq 1$ with $\tau_l \simeq \tau_{l+p}$. Figure 1 separates the initial segment of this chain from the repeated segment. The key point is that periodicity at the repeated class forces every pullback of that representation to remain irreducible, so that periodicity reaches the starting class.

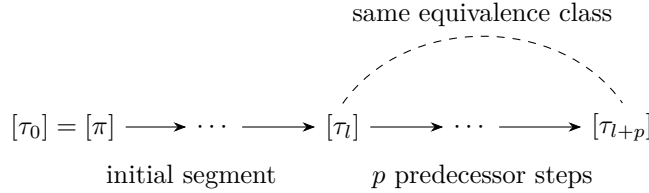


Figure 1: A chain inside one finite set D . An arrow from $[\tau_i]$ to $[\tau_{i+1}]$ means $\tau_i \prec \tau_{i+1} \circ \theta$. Repetition gives $\tau_l \simeq \tau_l \circ \theta^p$ by equality of dimensions. The proof then shows that each pullback of τ_l is irreducible, allowing this periodicity to pass back to π . The initial segment can be empty.

Lemma 5.4. *Let $\theta : \Gamma \rightarrow \Gamma$ be a homomorphism, and let D be a finite set of equivalence classes of nonzero finite-dimensional irreducible unitary representations of Γ . Suppose that every $[\pi] \in D$ has $[\pi'] \in D$ with $\pi \prec \pi' \circ \theta$. Then for every $[\pi] \in D$ there is an integer $p \geq 1$ such that $\pi \simeq \pi \circ \theta^p$.*

Proof. Starting from $\tau_0 = \pi$, choose a chain in D with $\tau_i \prec \tau_{i+1} \circ \theta$. Finiteness gives $\tau_l \simeq \tau_{l+p}$ for some $l \geq 0$ and $p \geq 1$. Composing the inclusions along this segment yields

$$\tau_l \prec \tau_{l+p} \circ \theta^p \simeq \tau_l \circ \theta^p.$$

The first and last representations have the same dimension, so

$$\tau_l \simeq \tau_l \circ \theta^p. \tag{16}$$

It follows by iteration that $\tau_l \circ \theta^{kp} \simeq \tau_l$ for every integer $k \geq 0$.

For every $j \geq 0$, the representation $\tau_l \circ \theta^j$ is irreducible. Indeed, if it had a nonzero proper invariant subspace, that subspace would remain invariant for $\tau_l \circ \theta^{kp}$ whenever $kp \geq j$, because

$\theta^{kp}(\Gamma) \subseteq \theta^j(\Gamma)$. This contradicts the irreducibility of $\tau_l \circ \theta^{kp} \simeq \tau_l$. The initial segment of the chain gives $\pi \prec \tau_l \circ \theta^l$. Its right-hand side is irreducible, so $\pi \simeq \tau_l \circ \theta^l$. Composing (16) with θ^l therefore gives

$$\pi \circ \theta^p \simeq \tau_l \circ \theta^{l+p} \simeq \tau_l \circ \theta^l \simeq \pi.$$

□

Choose one index n in the intersection of the ω -large sets from Lemmas 5.2 and 5.3. Apply Lemma 5.4 to D_n and π_* . It gives $\pi_* \simeq \pi_* \circ \theta^p$ for some $p \geq 1$. But π_* acts nontrivially on the translations $\mathbb{Z}^3$, whereas $\pi_* \circ \theta^p$ acts trivially on them: θ^p multiplies these translations by 2^p , and π_* factors through reduction modulo 2. This is the required contradiction.

The argument used only that B is nonzero, unital and nuclear, and that ω is free. The group G and the algebra $C^*(G)$ are independent of these choices. Thus the same algebra has no unital embedding into B^ω for any such B and any free ω . Taking $B = \mathcal{O}_2$ proves the stated negative answer to Kirchberg's embedding problem.

6 Consequences for existential closure and nuclear approximation

We spell out the model-theoretic consequences of the negative answer to Kirchberg's embedding problem. In the unital C^* -algebra language, quantifier-free formulas are continuous combinations of norms of $*$ -polynomials; variables range over prescribed norm balls. A unital algebra E is *existentially closed* if, for every unital inclusion $E \subseteq F$, every finite parameter tuple a from E , and every quantifier-free formula $\varphi(a, y)$ in a finite tuple y ,

$$\inf_{y \in E} \varphi(a, y) = \inf_{y \in F} \varphi(a, y),$$

where y ranges over the corresponding products of prescribed norm balls. Thus an extension cannot improve the infimum of a finite quantifier-free test over E [6, Section 2.1].

A parameter-free finite condition consists of finitely many strict inequalities $\varphi_j(x) < r_j$ with quantifier-free φ_j . It is *satisfiable* when some tuple in a unital C^* -algebra satisfies them. The last assertion below makes the failure of good nuclear witnesses explicit as a positive lower bound on the approximation error of every finite-dimensional completely positive contractive factorization of every realizing tuple.

The KEP equivalences of Goldbring–Sinclair [6, Proposition 3.2 and Theorems 3.3 and 3.7] give these direct consequences of Theorem 1.1.

Corollary 6.1 (Consequences of negative KEP). *Neither $\mathcal{O}_2$ nor any exact or nuclear unital C^* -algebra is existentially closed among unital C^* -algebra extensions.*

For every free ultrafilter ω on $\mathbb{N}$, there are a separable unital C^ -algebra D and $\alpha \in \text{Aut}(D)$ such that D embeds unittally into $\mathcal{O}_2^\omega$ but $D \rtimes_\alpha \mathbb{Z}$ does not embed unittally there.*

Some satisfiable finite condition $p(x) = \{\varphi_j(x) < r_j : 1 \leq j \leq s\}$ in the unital C^ -algebra language, with $x = (x_1, \dots, x_m)$, quantifier-free φ_j and $r_j > 0$, lacks good nuclear witnesses: for some $\varepsilon_0 > 0$, every realization $a = (a_i)_{i=1}^m$ of p in every unital C^* -algebra E satisfies $\max_{1 \leq i \leq m} \|\psi\phi(a_i) - a_i\| \geq \varepsilon_0$ for every finite-dimensional C^* -algebra F and all completely positive contractive maps $\phi : E \rightarrow F$, $\psi : F \rightarrow E$, not necessarily unital.*

Proof. For each fixed free ω , Theorem 1.1 with $B = \mathcal{O}_2$ negates unital KEP. The existential-closure claims follow from [6, Theorem 3.3], using that nuclearity implies exactness. Proposition 3.2 of the same paper, with its preceding construction for separable unital algebras, gives the crossed-product claim; Definition 3.6 and Theorem 3.7 give the finite-condition claim. These two witnesses are existential, not identified by our explicit group construction. □

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